STRUCTURAL ANALYSIS 1 (INS2512) - LESSON 1 -

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SOURCE BOOKS OF THE COURSE

- İbrahim Ekiz (2010), *YAPI STATİĞİ 1-İzostatik Sistemler*, Birsen Yayınevi
- Faruk Karadoğan, Sumru Pala, Ercan Yüksel, Yavuz Durgun (2011), *YAPI MÜHENDİSLİĞİNE GİRİŞ – YAPISAL ÇÖZÜMLEME*, Birsen Yayınevi
- İrdesel Göğüş ve Bilge Doran, Yapı Statiği 1, YTÜ Yayınları
- R. C. Hibbeler (2014), *STRUCTURAL ANALYSIS 9th EDITION*, Prentice Hall

REMINDERS

- Evaluation System:

1) First Mid-Term Exam (Percentage of grade is 30%)

2) Second Mid-Term Exam (Percentage of grade is 30%)

3) Final Exam (Percentage of grade is 40%)

- Attendance:

It is crucial to attend the lessons in order to understand the topics and be successful in this course!!

INTRODUCTION

What is a structure?

A *structure* refers to a system of connected parts used to support a load.

Common examples of civil engineering structures?

- Buildings
- Bridges
- Towers
- Tunnels
- Dams
- Off-shore structures
- Industrial Structures
- Etc.

When designing a structure, the engineer must account for its;

- safety
- esthetics
- serviceability

while considering also economic and environmental constraints.

This requires several independent studies of different solutions before reaching a final judgement on the structural form.

The design process is both creative and technical, and it requires a fundamental knowledge of material behavior and the laws of mechanics governing material response.

Once a preliminary design of a structure is made, the structure must be analyzed to ensure that it has the required stiffness and strength.

For this purpose,

- certain idealizations must be made as how structural elements are supported and connected to each other.
- The loadings are determined according to the codes
- Then the forces in the elements and their displacements are calculated by means of the theory of *structural analysis.*
- The results of the first analysis can be used to re-design the structure

The results of the first analysis can be used to re-design the structure to achieve an optimum design.

In summary, structural design follows a series of successive aproximations in which every cycle requires a structural analysis.

STRUCTURAL SYSTEMS

Common examples of civil engineering structures?

- Buildings
- Bridges
- Towers
- Tunnels
- Dams
- Off-shore structures
- Industrial Structures
- Etc.

A structure consists of various structural elements:

Tie Rods: Structural members subjected to a tensile or compression force are often referred to as *tie rods* or *bracing struts*.



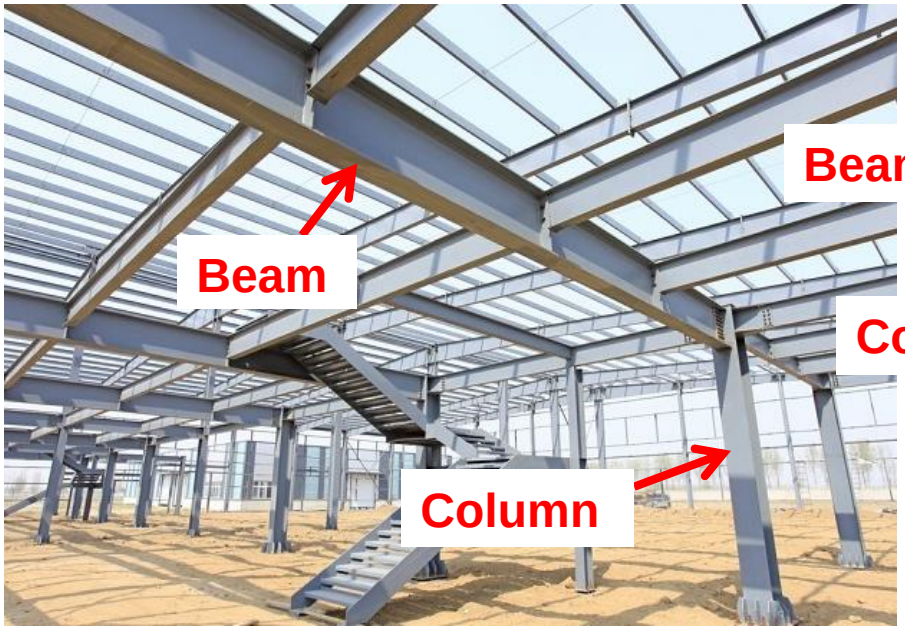


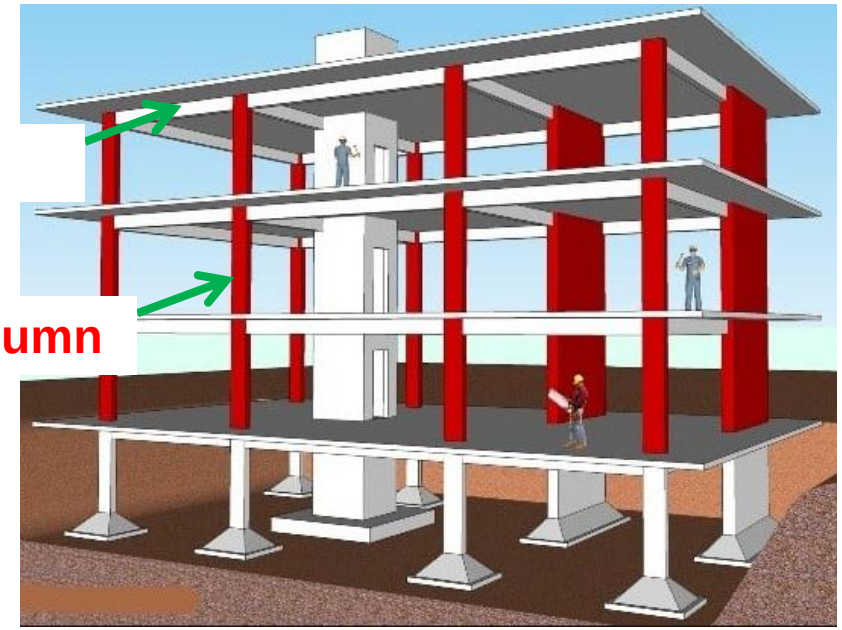




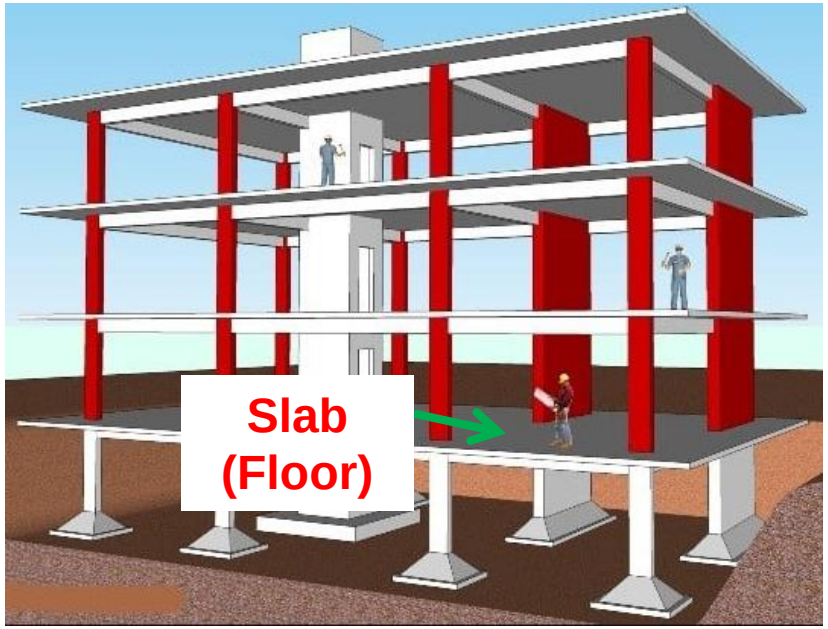
Beams: Beams are usually straight horizontal members used primarily to carry vertical loads.

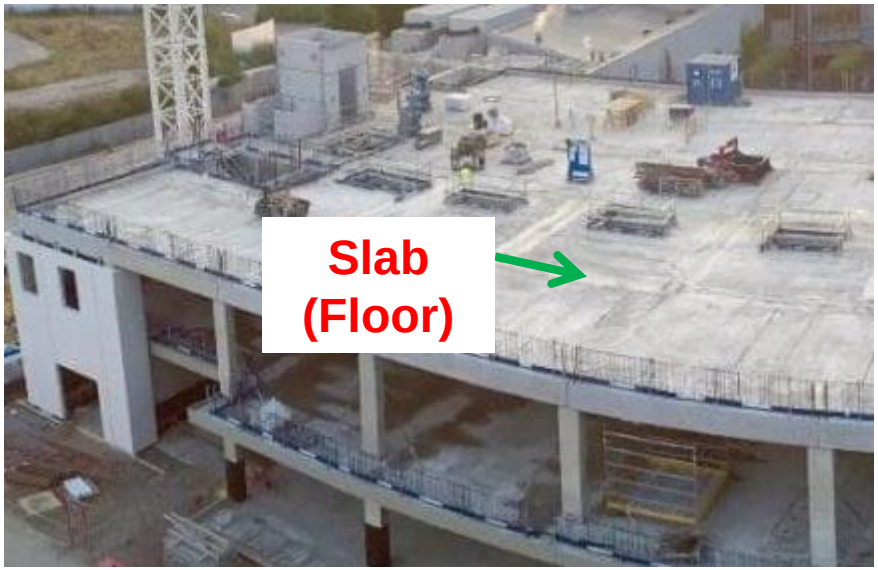
Columns: Members that are generally vertical and resist axial compressive load and bending moment are referred to as columns.





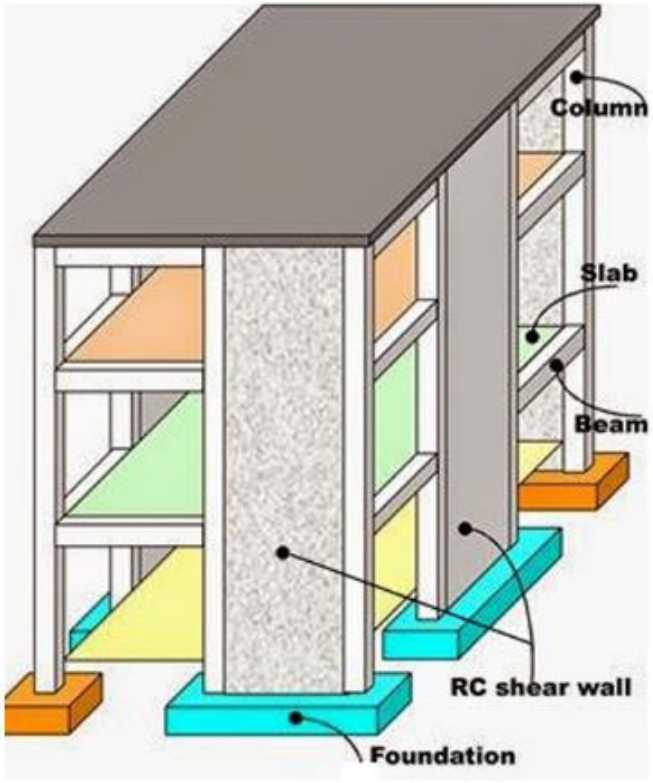
Slabs: Slab is an important structural element which is constructed to create flat and useful surfaces such as floors, roofs, and ceilings. It is a horizontal structural component, with top and bottom surfaces parallel or near so.





Shear wall: Shear wall is a structural member in a reinforced concrete framed structure to resist lateral forces such as wind or seismic forces. Shear walls are generally used in high-rise buildings subject to lateral wind and seismic forces.





Structural Elements

There are more different structural elements except the given ones. These are the examples to understand the members of a structure.

A structure must be analytically modelled to estimate the internal forces of the elements and their displacements. Thus, some idealizations must be done to analytically model the structural elements.

The structural elements are classified into three groups according to their dimensions in an analytical model:

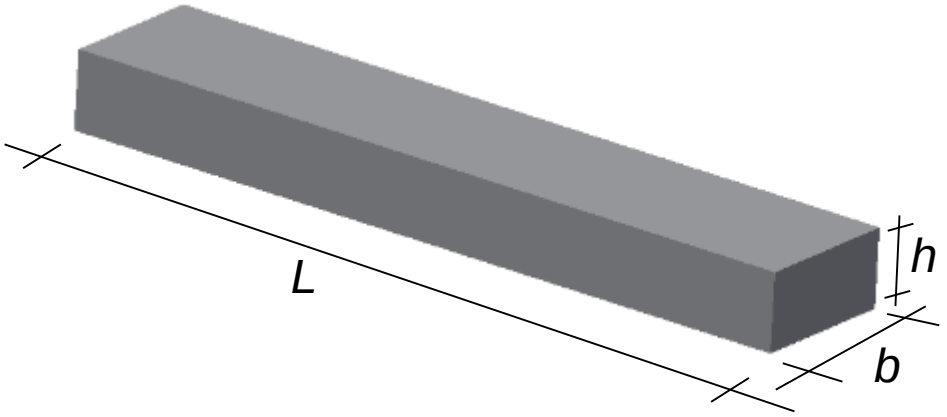
1. Linear (1D) structural elements
2. Surface (2D) structural elements
3. 3D structural elements

1. Linear (1D) Structural Elements

Structural elements having two dimensions which are small compared to the third one are referred to as linear (1D) structural elements.

Example:

- Tie Rod
- Beam
- Column



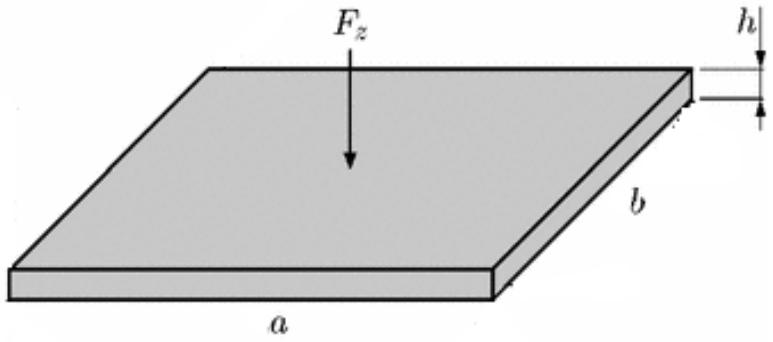
b, h << *L*

2. Surface (2D) Structural Elements

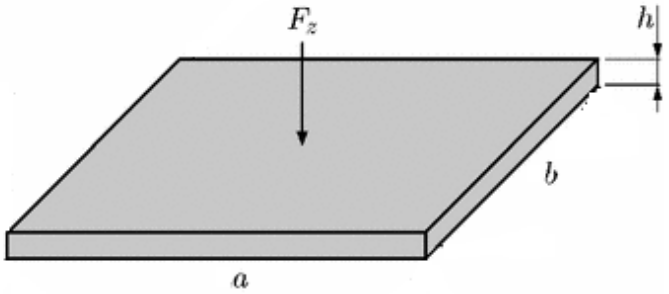
Structural elements having two dimensions in similar orders and a third dimension which is small compared to the other two dimensions are referred to as surface (2D) structural elements.

Example:

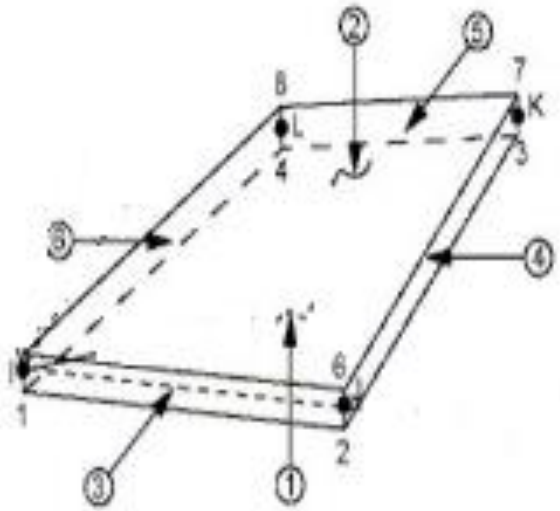
- Slab
- Shear wall



$h << a, b$



If the element is planar and the forces are normal to the plane, then the element is refered to as plate.



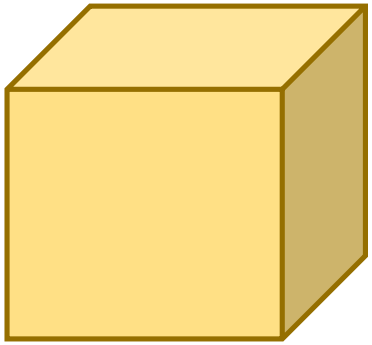
If it is a curved element and the loads are in general condition, it is referred to as shell.

3. 3D Structural Elements

Structural elements with all dimensions in the same order are referred to as 3D structural elements.

Example:

- Concrete blocks



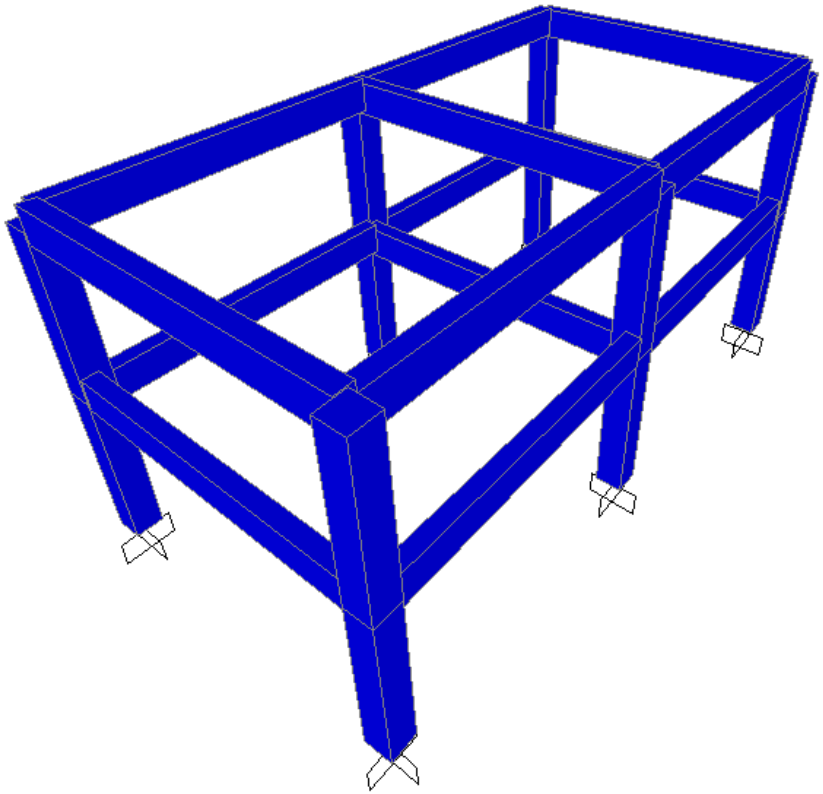
In the context of this course, structures composed of 1D elements such as beams and columns are considered.

Idealization

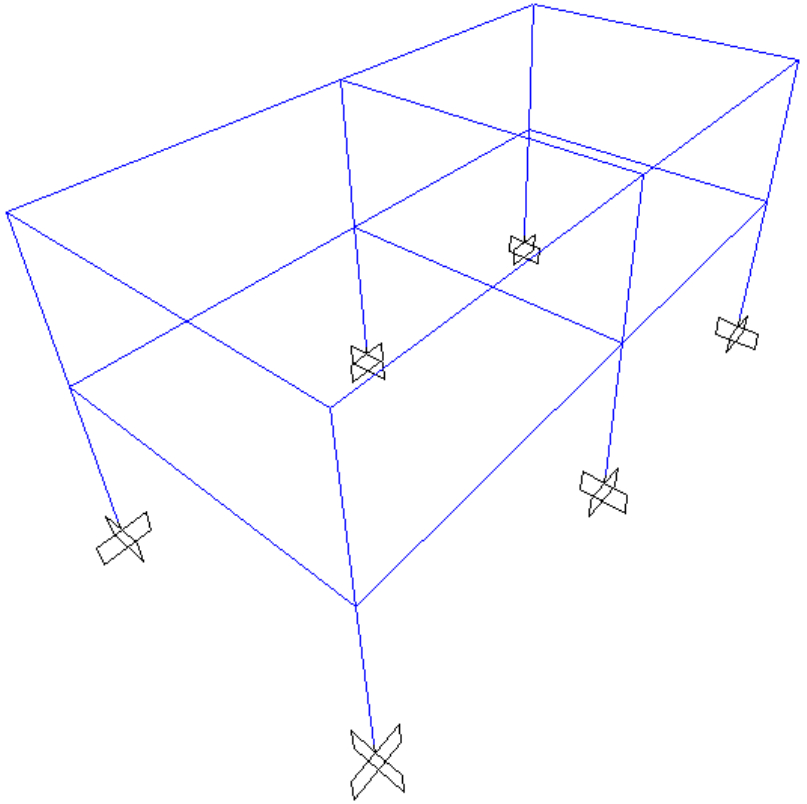
Actual structures are too complex to be analyzed as they are, so they are idealized by means of some simplifications.

Structural elements of an actual structure composed of beams and columns are generally modelled as 1D elements since their cross-sectional dimensions are too small compared to their length.

These structures are referred to as frame structures.

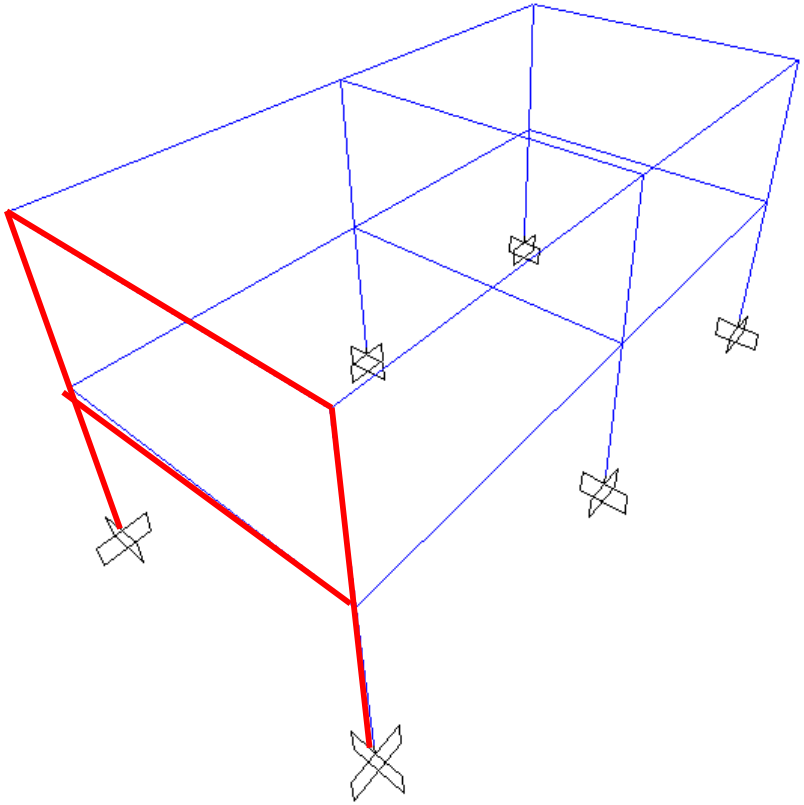


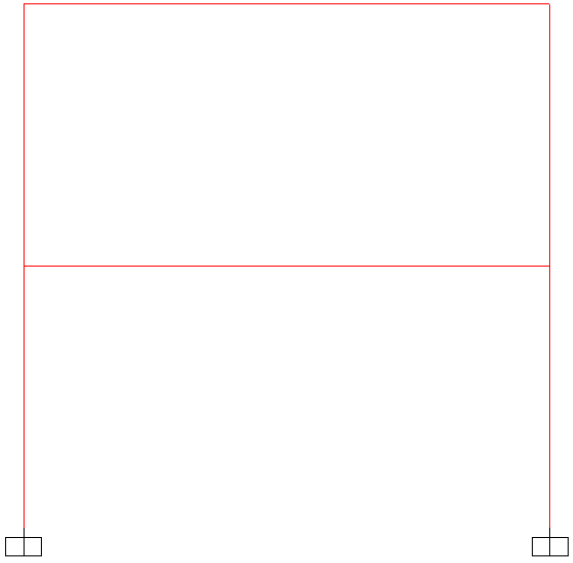
3D structure with beams and columns



3D structure with 1D elements

Under certain conditions, 3D frame structures can be further simplified by splitting it into a series of 2D frames.



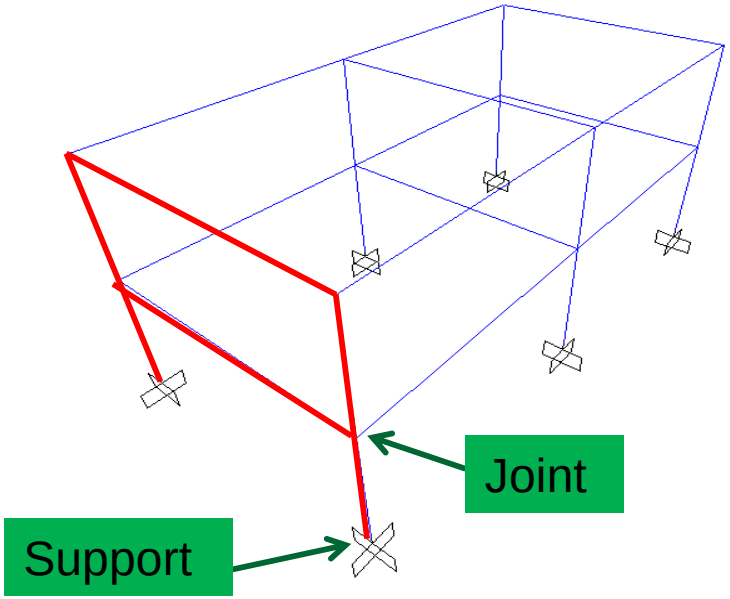


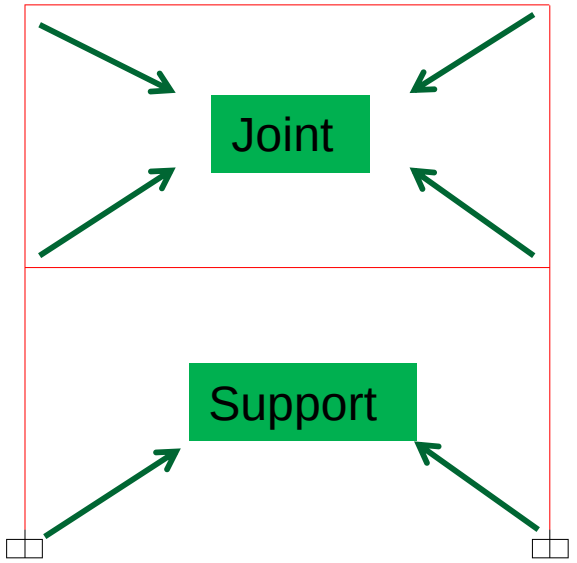
2D structure with 1D elements

Supports and Joints

A structural element tranfering the loads from a structural system to either another structural system or soil is referred to as support.

The connection of structural elements is referred to as joint.



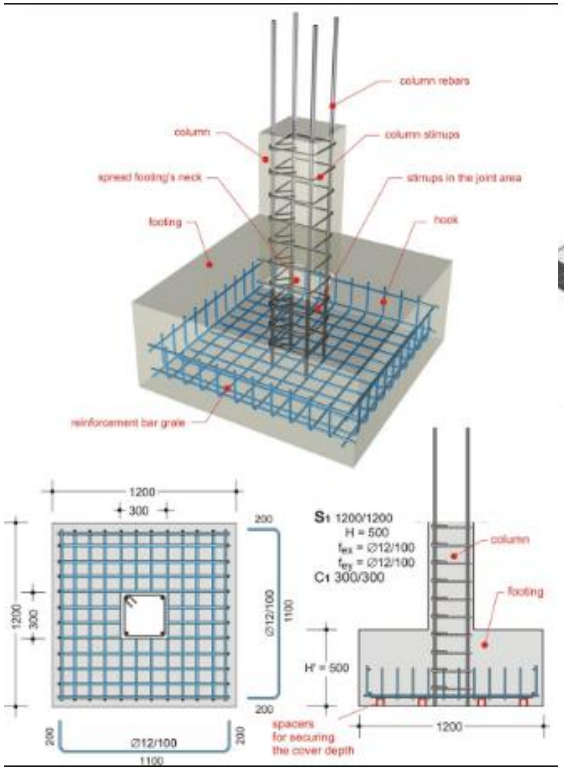


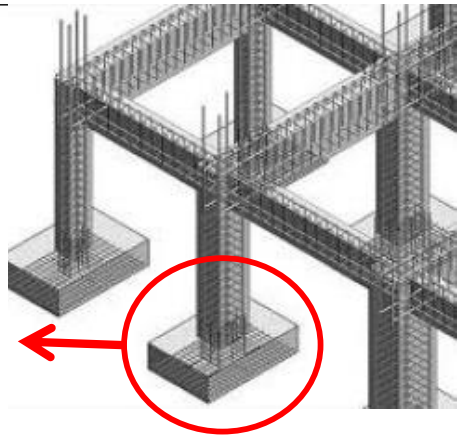
Supports

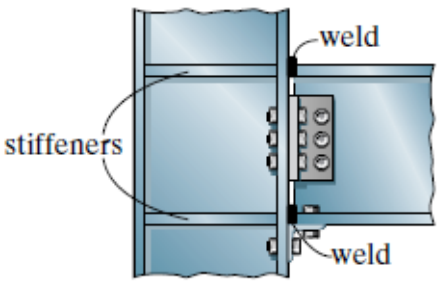
Supports are classified based on the type of reaction forces and the degree of freedoms they have. The common types of supports are

- Fixed support
- Pin support
- Roller support
- Slider support
- Elastic support
- Short link

Fixed Support





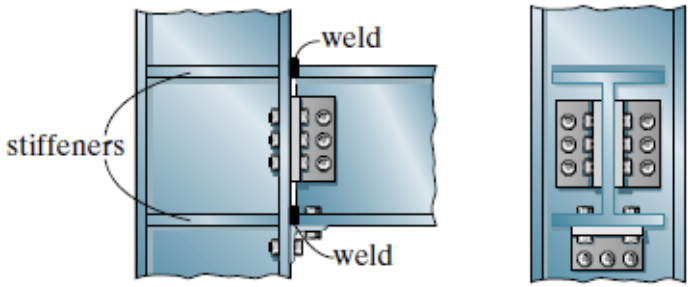


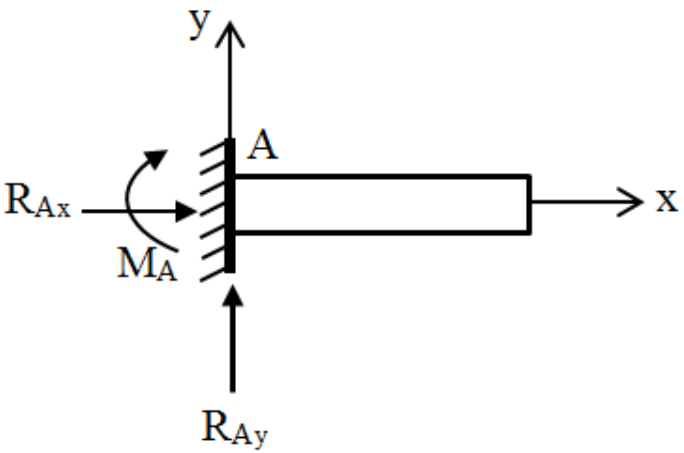




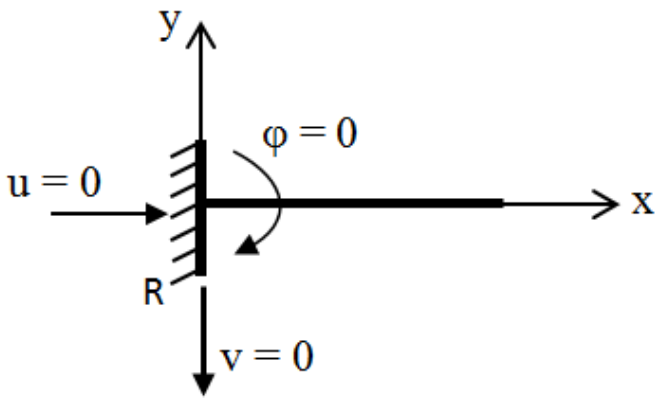
A support that connects a structural element to an infinitely rigid body or another structural system by restraining all degrees of freedom (cannot rotate and displace) is referred to as a fixed support.

Fixed Support





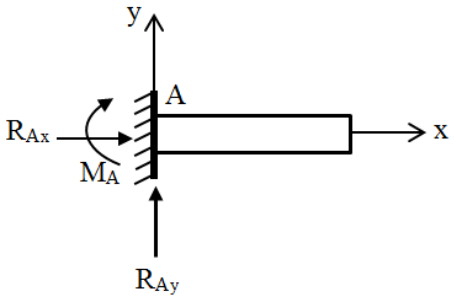
Support reaction



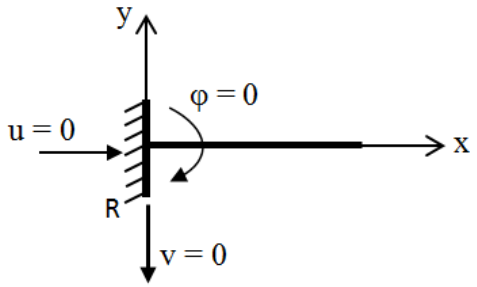
Displacements

Planer (2D) fixed support

Fixed Support



Support reaction



Displacements

Planer (2D) fixed support

- u = 0 (the displacement parallel to the axis of structural element)
- v = 0 (the displacement perpendcular to the axis of structural element)
- φ = 0 (the rotation about the axis normal to the structural element)

Since these displacements are restrained, there exist support reactions denoted as R_{Ax}, R_{Ay} and M_A.

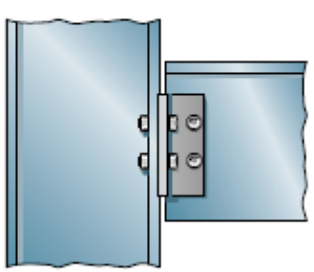
- there are three unknowns (or support reactions)
- the number of degree of freedom is zero (u = 0, v = 0, φ = 0)

Pin Support

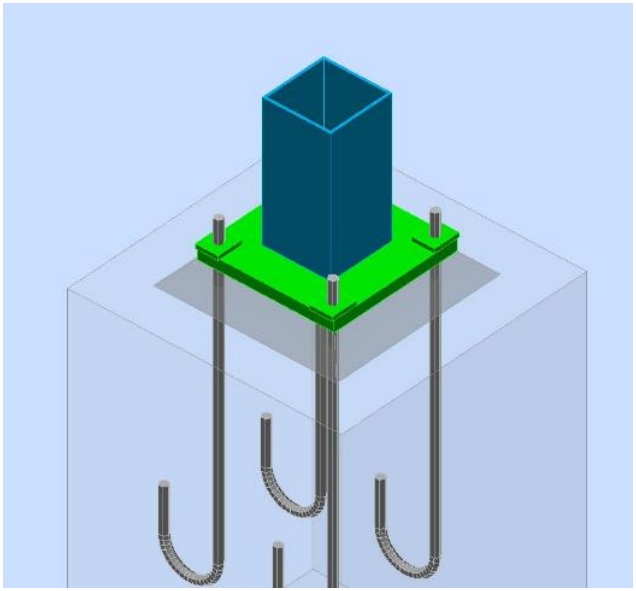








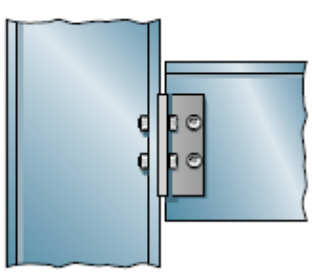




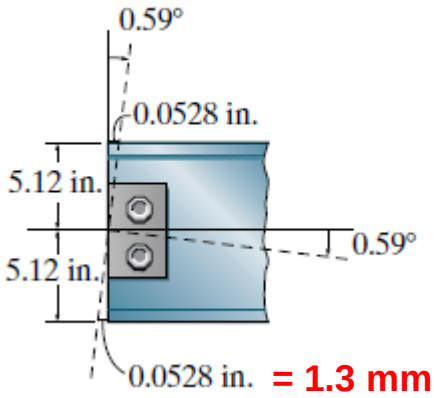
A support that allows the rotation of the structural element but restrains the translational displacements is referred to as a pin support.

Pin Support

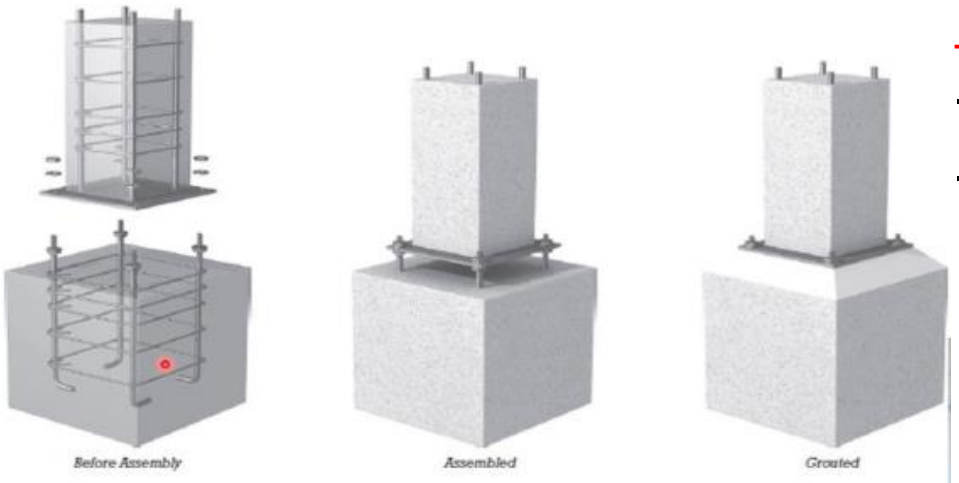








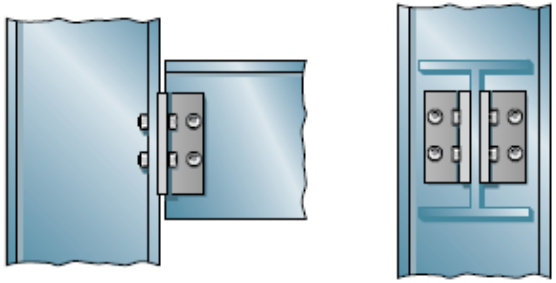
= 1.3 mm

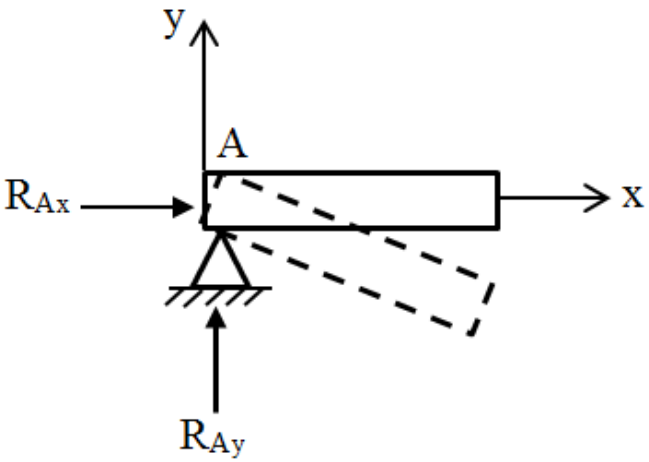


Pin support:

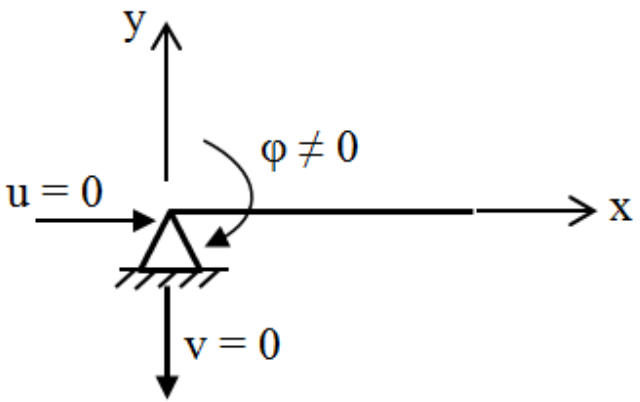
- Allows rotation
- Restrains translational displacement

Pin Support





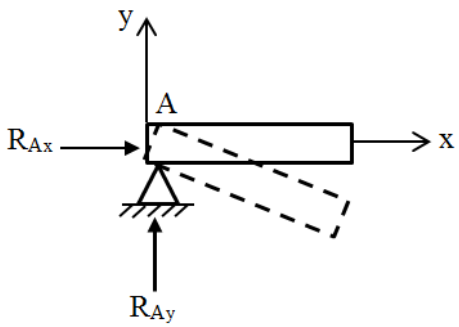
Support reaction



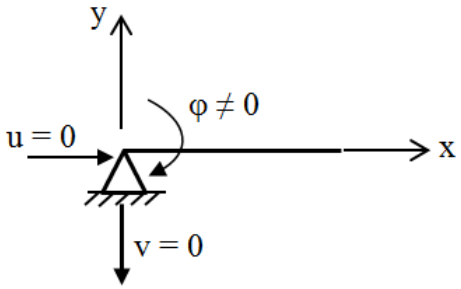
Displacements

Planer (2D) pin support

Pin Support



Support reaction



Displacements

Planer (2D) pin support

- u = 0 (the displacement parallel to the axis of structural element)
- v = 0 (the displacement perpendicular to the axis of structural element)
- φ ≠ 0 (the rotation about the axis normal to the structural element)

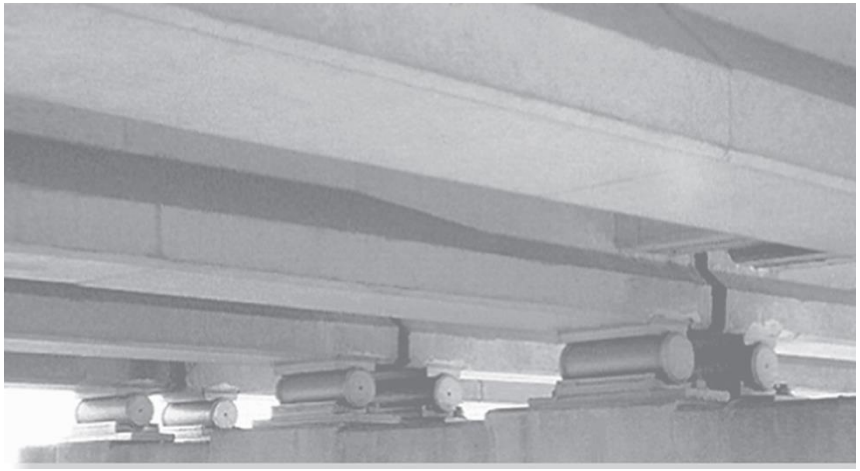
There exist support reactions denoted as R_{Ax}, R_{Ay} in the directions of x and y, since translational displacements are restrained in these directions. **The moment M_A is equal to zero, since rotation is allowed (φ ≠ 0) .**

- there are two unknowns (or support reactions)
- the number of degree of freedom is one (u = 0, v = 0, φ ≠ 0)

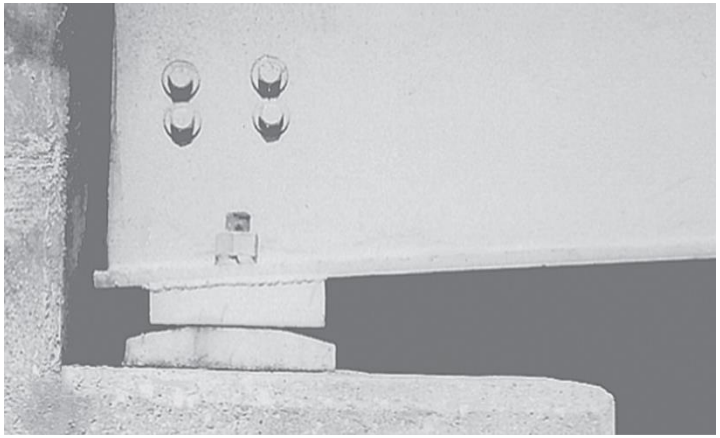
Roller Support



Notice that the deck of this concrete bridge is made so that one section can be considered roller supported on the other section.



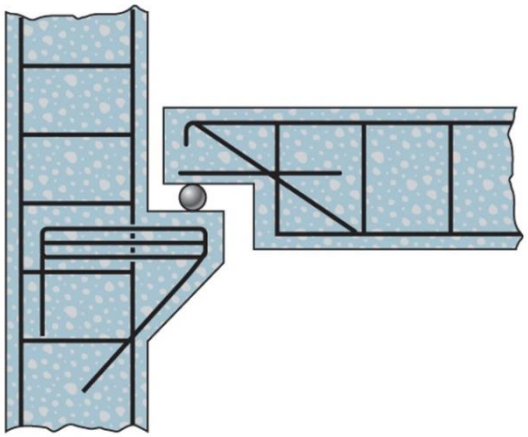
Rollers and associated bearing pads are used to support the prestressed concrete girders of a highway bridge.

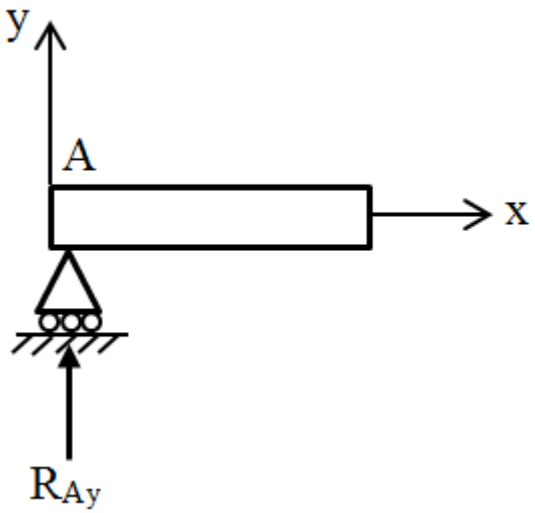


A typical rocker support used for a bridge girder.

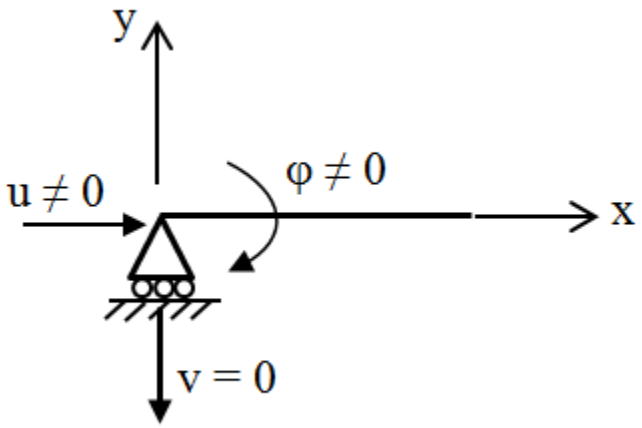
A support that allows the rotation of the structural element and the translational displacement in one direction in 2D-systems is referred to as a roller support.

Roller Support





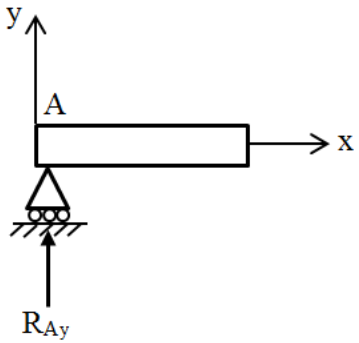
Support reaction



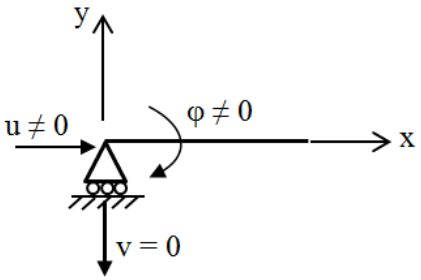
Displacements

Planer (2D) roller support

Roller Support



Support reaction



Displacements

Planer (2D) roller support

- u ≠ 0 (the displacement parallel to the axis of structural element)
- v = 0 (the displacement perpendicular to the axis of structural element)
- φ ≠ 0 (the rotation about the axis normal to the structural element)

There exist a support reaction denoted R_{Ay} in the directions of v, and the other support reactions R_{Ax} and M_A are equal to zero.

The moment M_A is equal to zero, since rotation is allowed (φ ≠ 0) .

- there is one unknown (or support reaction)
- the number of degree of freedom is two (u ≠ 0, v = 0, φ ≠ 0

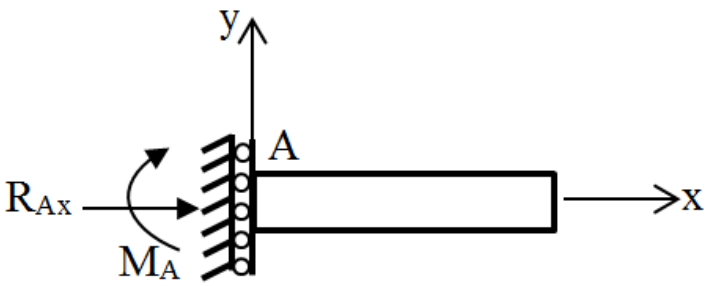
Slider Support



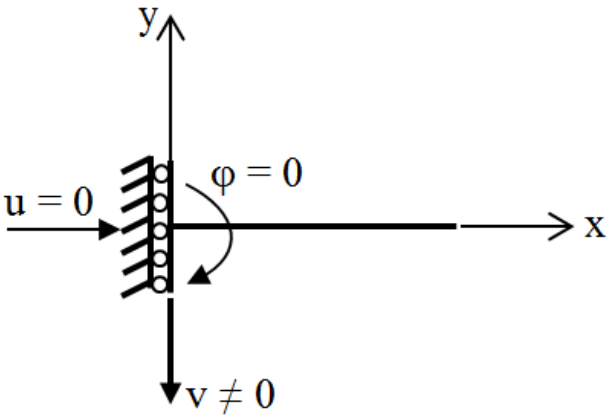


A support that allows the translational displacement of the structural element in a certain direction while restraining the other translational direction and the rotational displacement is referred to as a slider support.

Slider Support



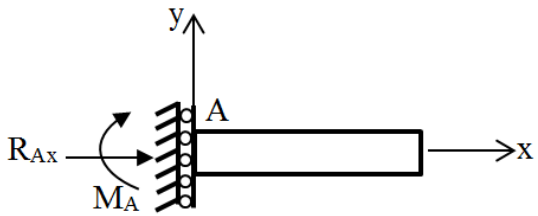
Support reaction



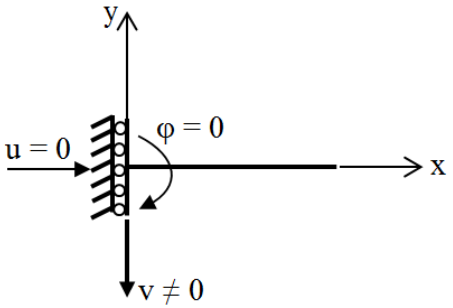
Displacements

Planer (2D) slider support

Slider Support



Support reaction



Displacements

Planer (2D) slider support

- u = 0 (the displacement parallel to the axis of structural element)
- v ≠ 0 (the displacement perpendicular to the axis of structural element)
- φ = 0 (the rotation about the axis normal to the structural element)

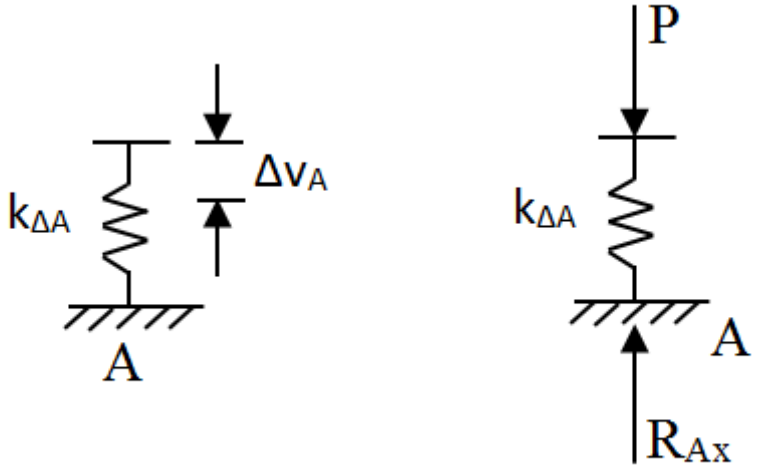
There exist two support reactions, a force in the direction of u and a moment in the direction of φ. The support reaction in the direction of v is zero.

- there are two unknowns (or support reactions)
- the number of degree of freedom is one (u = 0, v ≠ 0, φ = 0)

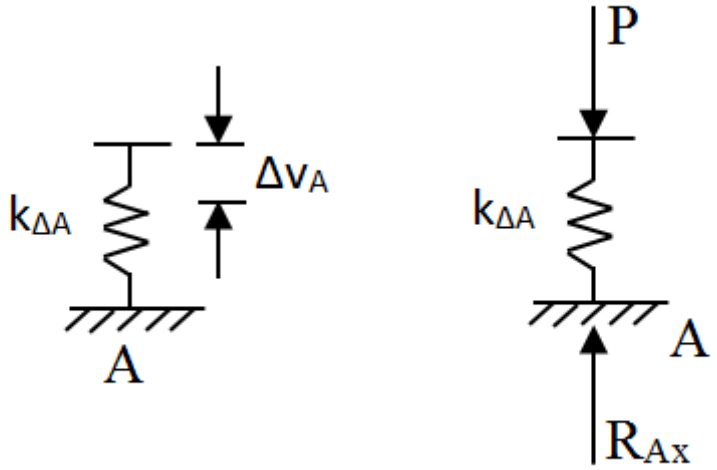
Elastic Support

i) Translational Elastic Support

A support that allows a limited amount of translational displacement in one direction while restraining translational displacement in the other direction and rotational displacement is referred to as a translational elastic support.



i) Translational Elastic Support



$$\Delta v_A = \frac{P}{k_{\Delta A}} = \frac{R_{Ay}}{k_{\Delta A}}$$

For example, a foundation on soft soil is modelled as elastic support.

Due to the load on the foundation, there occurs elastic settlement of soil, which is modelled by means of a translational spring with a stiffness k_Δ.

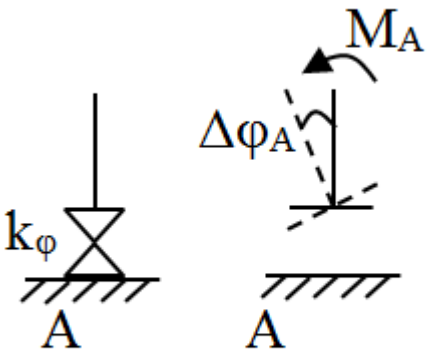
k_Δ is a force that results a unit displacement (Δv_A = 1) in the spring.

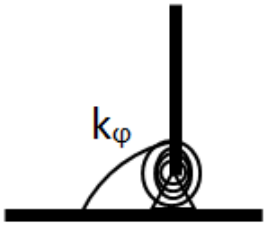
Elastic Support

ii) Rotational Elastic Support

A support that allows rotational displacement proportional to the moment while restraining translational displacements parallel and perpendicular to the axis of structural element in one direction while restraining translational displacement is referred to as a rotational elastic support.

k_φ is rotational stiffness of a spring and k_φ is a moment results unit rotation ($\Delta\varphi_A = 1$).





$$\Delta\varphi_A = \frac{M_A}{k_{\varphi A}}$$

Elastic Support

iii) Rotational and Translational Elastic Support

In an actual foundation, both rotational and translational displacements occure together on a soft soil.

A support that allows both translational and rotational displacements proportional to the support reactions is referred to as a rotational and translational elastic support.

$k_{\Delta u}$ and $k_{\Delta v}$ are two spring stiffness constants related to translational displacements.

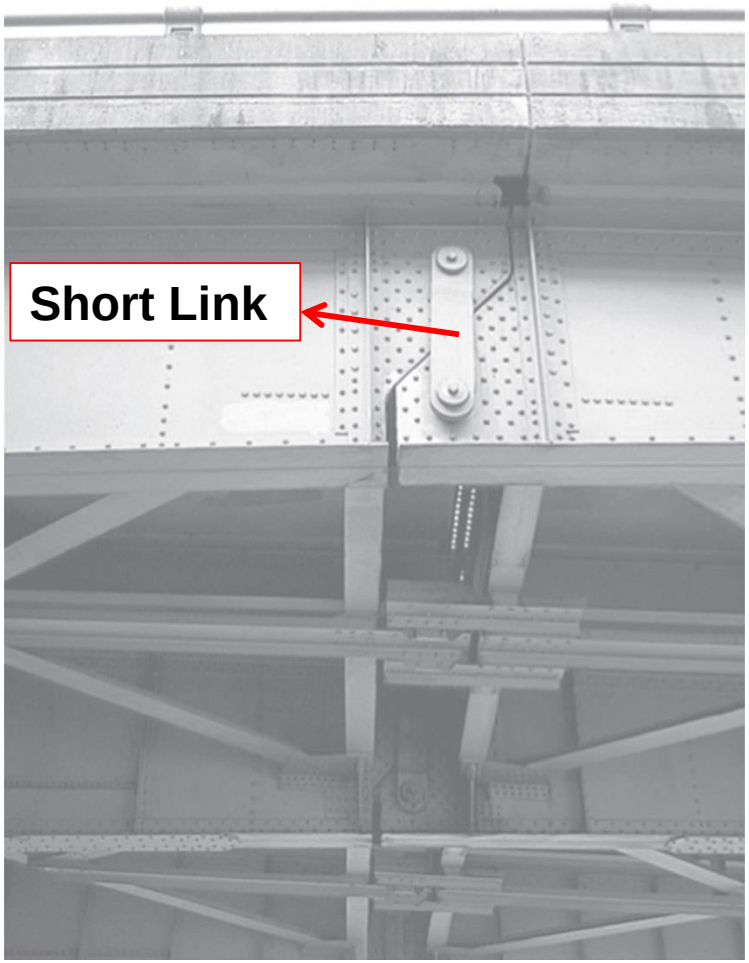
k_{φ} is spring stiffness constant related to rotational displacement.

Thus;

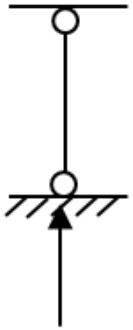
$$\Delta u = \frac{R_x}{k_{\Delta u}} \qquad \Delta v = \frac{R_y}{k_{\Delta v}} \qquad \Delta\varphi = \frac{M}{k_{\varphi}}$$

Short Link

A support made of a structural element pinned at both ends.



No force perpendicular to the axis of structural element acts and there is only one support reaction parallel to the axis of the structural element



Idealization of a short link

The short link is used to connect the two girders of the highway bridge and allow for thermal expansion of the deck

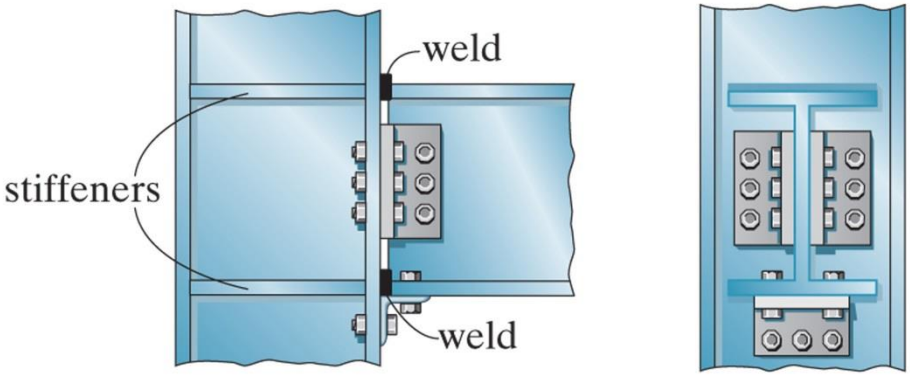
Joints

The connection of structural elements is referred to as joint. The joints are mainly classified as:

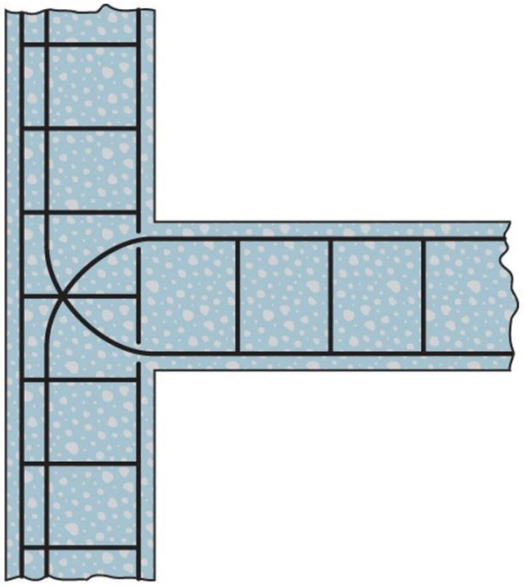
- Fixed joint
- Pin (hinge) joint
- Elastic joint

Fixed Joint

A fixed (rigid) joint provides a rigid connection. That means, structural elements framing into the same rigid joint cannot rotate and displace relative to each other. It allows the transfer of moments.

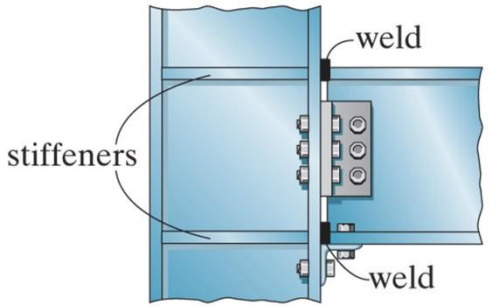


Steel structure connection

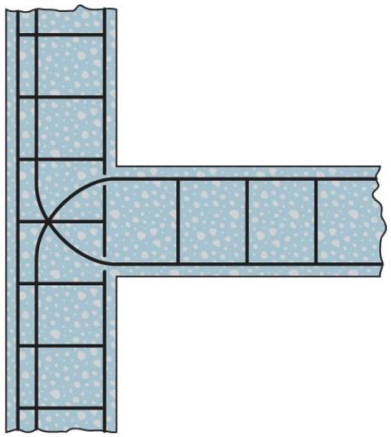


Reinforced concrete structure connection

Fixed Joint









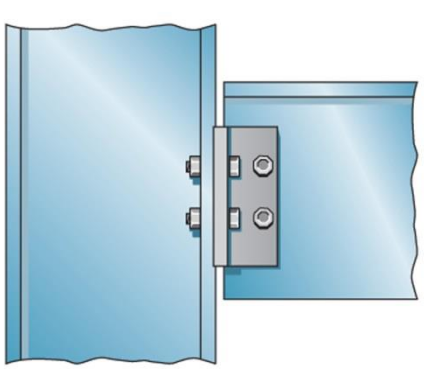
or



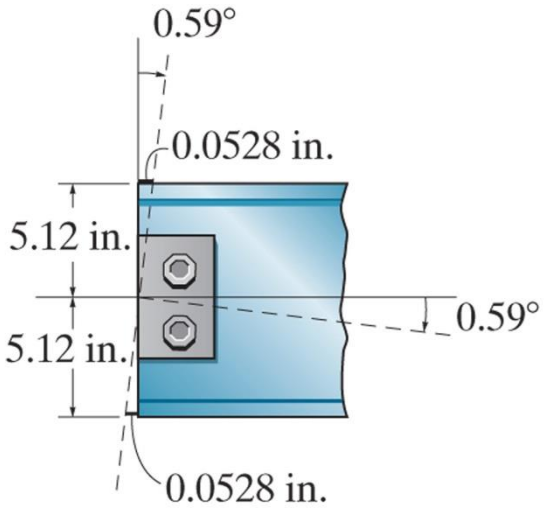
Idealization of a fixed joint

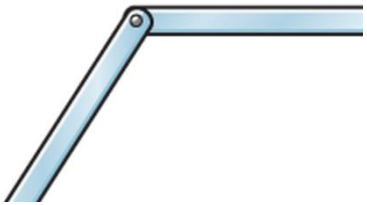
Pin Joint

A pin joint allows relative rotation of structural elements framing into the same pin joint, but does not allow the transfer of moment.









or

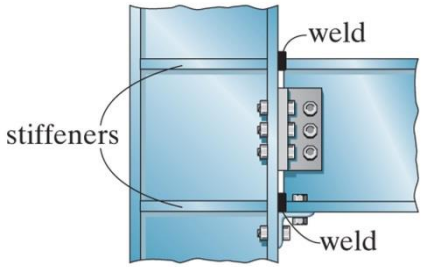


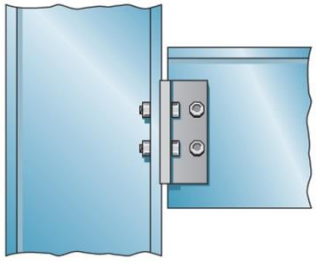
Idealization of a pin joint

Elastic Joint

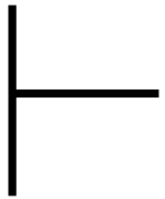
In an actual structure, there do not exist fixed or pin joints, but it can be assumed that some joints are fixed or pin joint.

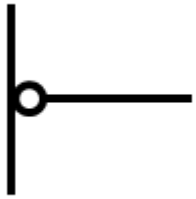
However, the most realistic approach is to model the connections as elastic joints with properly defined spring constants.



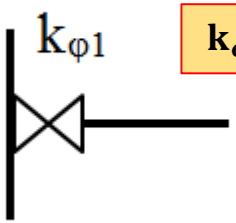


Connection of an actual structure

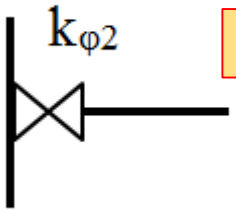




Idealization of the joint



$k_{\varphi 1}$ is very large



$k_{\varphi 2}$ is small

Realistic idealization of the joint with elastic connection

Summary of Supports
(R.C. Hibbeler, 2012)

TABLE 2–1 Supports for Coplanar Structures

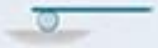
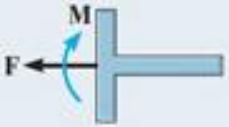
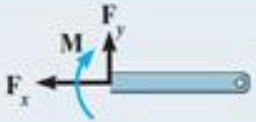
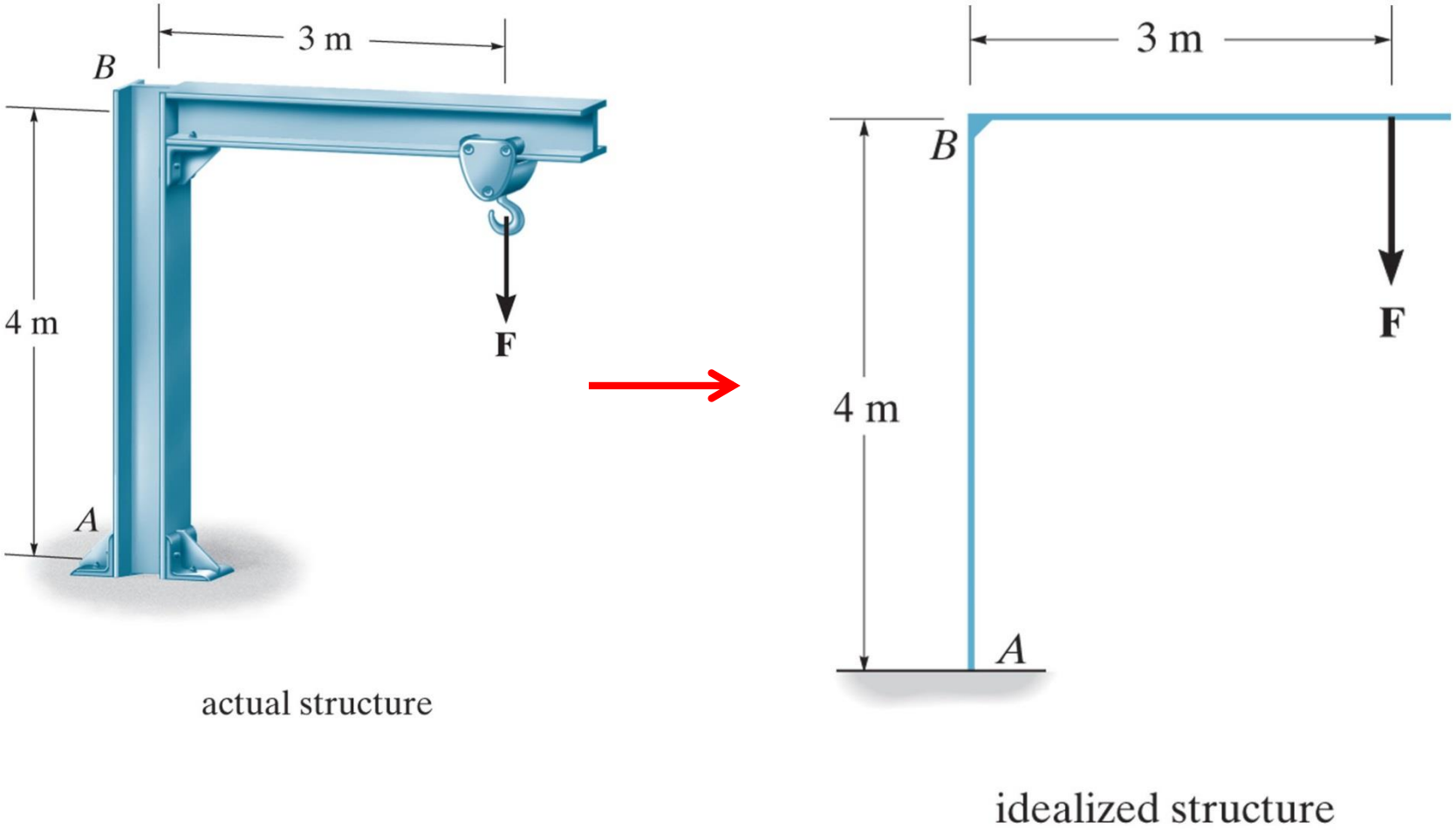
Type of Connection	Idealized Symbol	Reaction	Number of Unknowns
(1)  light cable  weightless link			One unknown. The reaction is a force that acts in the direction of the cable or link.
(2)  rollers   rocker	  		One unknown. The reaction is a force that acts perpendicular to the surface at the point of contact.
(3)  smooth contacting surface			One unknown. The reaction is a force that acts perpendicular to the surface at the point of contact.

TABLE 2–1 Supports for Coplanar Structures

Type of Connection	Idealized Symbol	Reaction	Number of Unknowns
(4)  smooth pin-connected collar			One unknown. The reaction is a force that acts perpendicular to the surface at the point of contact.
(5)  smooth pin or hinge			Two unknowns. The reactions are two force components.
(6)  slider  fixed-connected collar			Two unknowns. The reactions are a force and a moment.
(7)  fixed support			Three unknowns. The reactions are the moment and the two force components.

Example 1

Idealize the given system considering the structural analysis rules.



Example 2

Estimate the support reactions of the given system schematically.

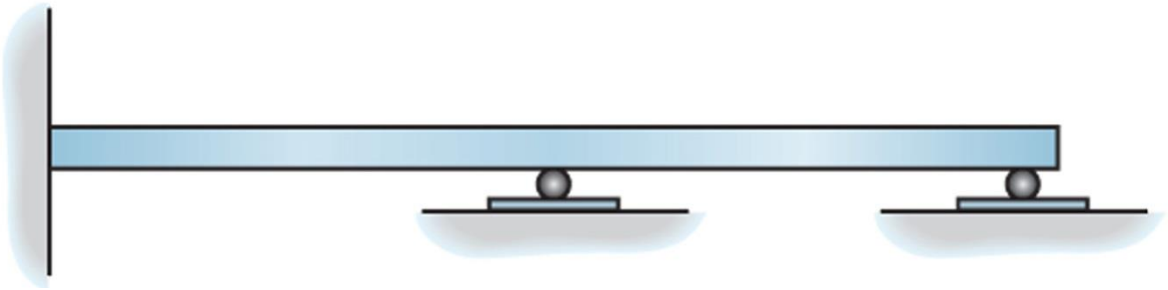


Solution:

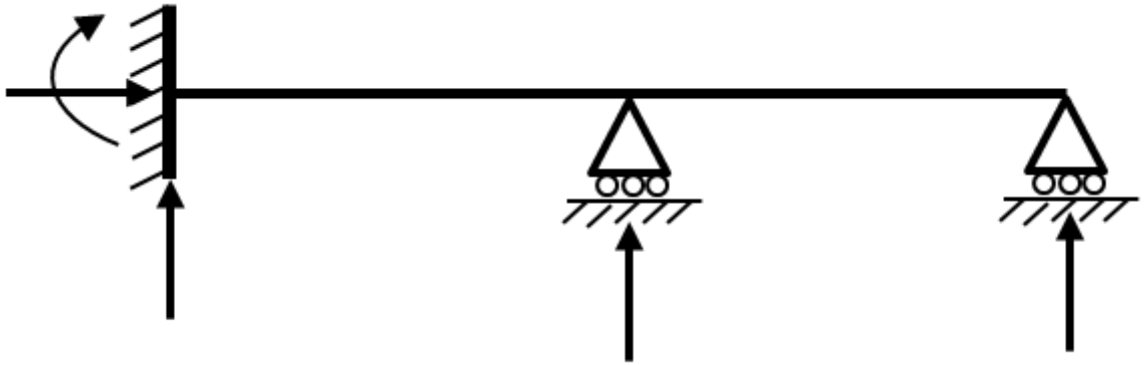


Example 3

Estimate the support reactions of the given system schematically.

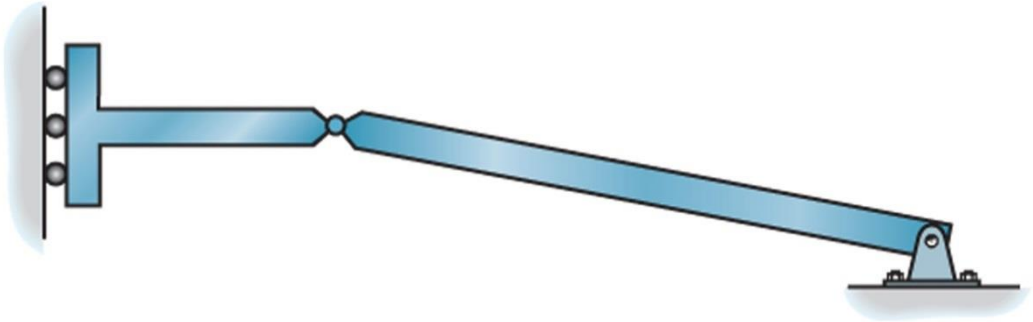


Solution:

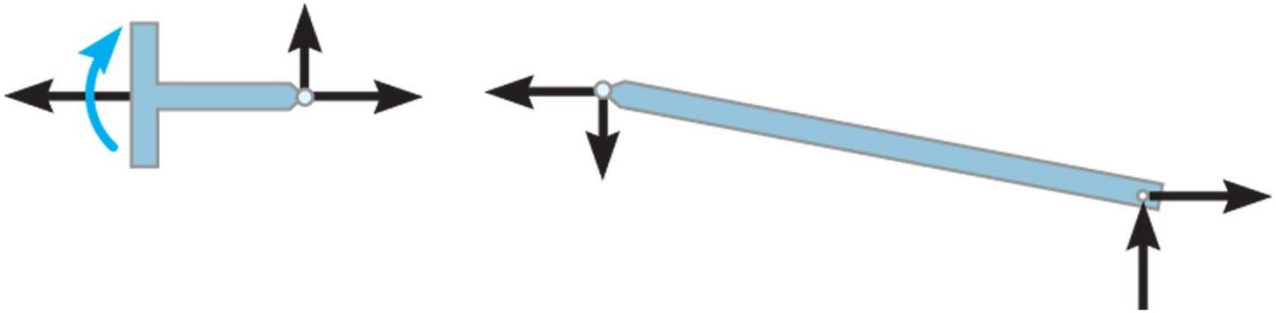


Example 4

Estimate the support and connection reactions of the given system schematically.

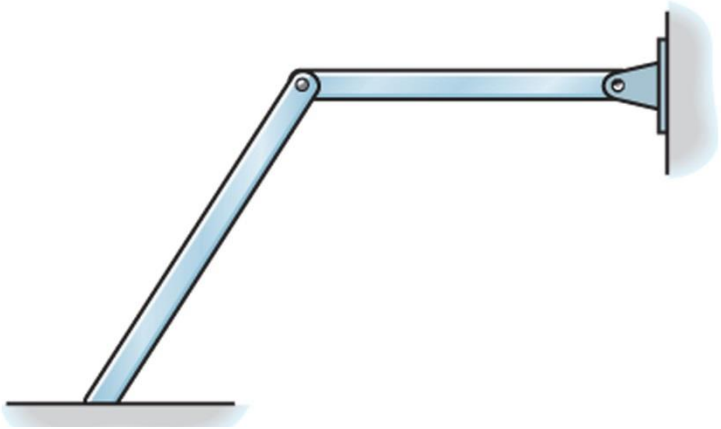


Solution:

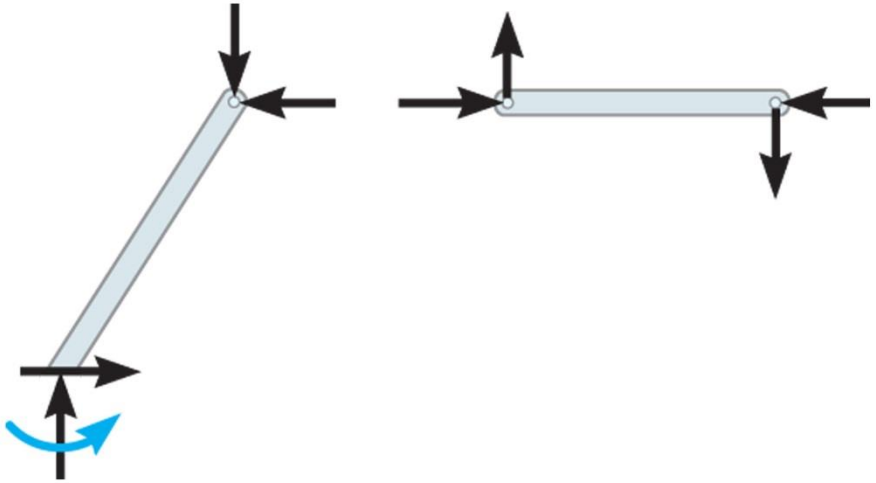


Example 5

Estimate the support and connection reactions of the given system schematically.

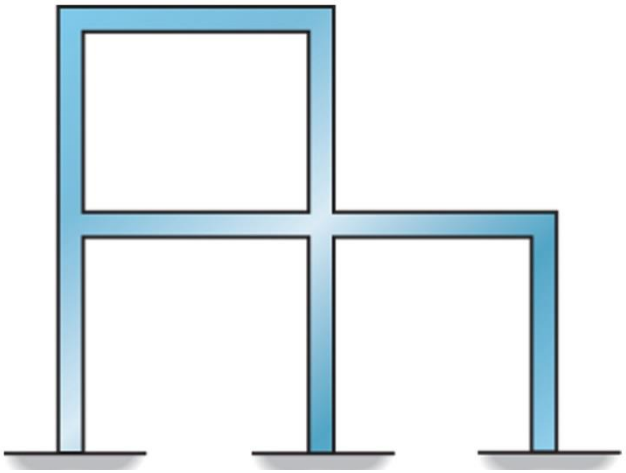


Solution:

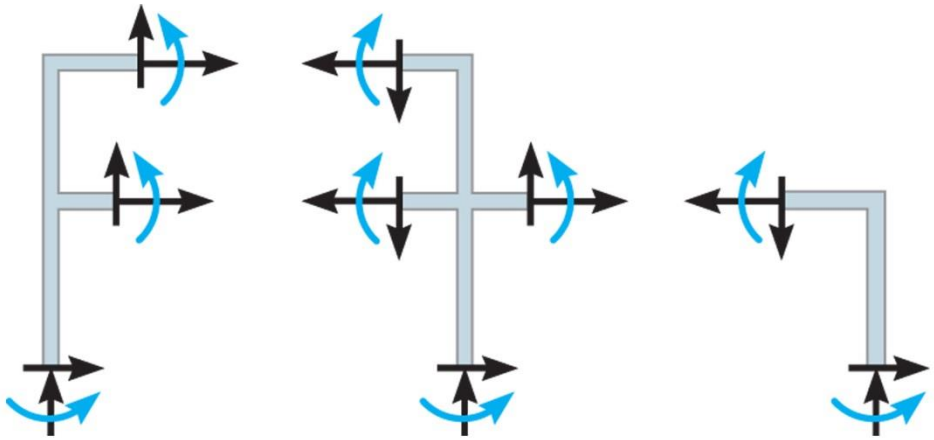


Example 6

Estimate the support and connection reactions of the given system schematically.

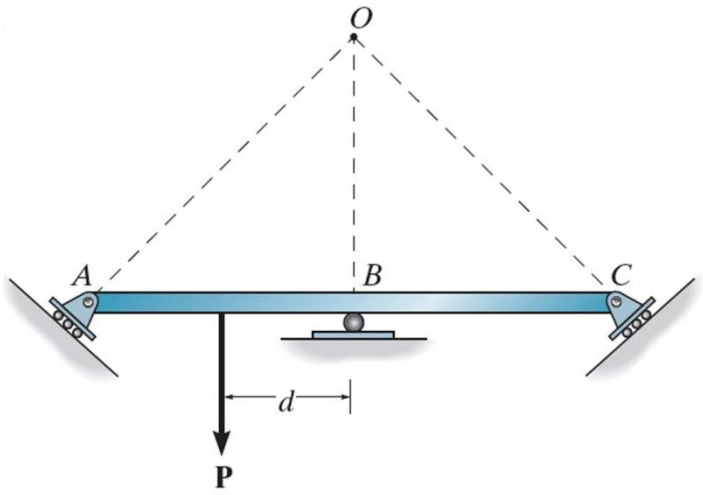


Solution:



Example 7

Estimate the support reactions of the given system schematically.



Solution:

